Joseph Elias

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Appointments

Binghamton University: Teaching Assistant	2025-
Binghamton University: Graduate Research Assistant	2024-2025
University of Pittsburgh: Bioacoustics Technician	2022-2024
Commonwealth University of Pennsylvania: Graduate Assistant	2022-2023
Keystone College: Adjunct Instructor	2021-2023

Education

Binghamton University — 2024-

- Ph.D. Ecology and Evolutionary Biology
- Research Advisor: Dr. Eliza Grames

Commonwealth University of Pennsylvania — 2022-2023

- M.S. Biology; GPA 3.91
- Thesis: *Microplastic Loading in Terrestrial Birds from Non-Lethal Sampling*
- Research Advisor: Dr. Clay Corbin

Keystone College — 2018-2021

- B.S. Environmental and Wildlife Biology; GPA 3.94
- Capstone title: *Seasonal Change in Water Quality and Species Composition of a Pennsylvania Tributary Trout Stream*
- Capstone Advisors: Drs. Valorie Titus & Robert Cook

Research Funding and Honors

\$3000	NATCHANGE Graduate Research Award	2026
\$2000	Shepherd Family Research Grant	2025
\$47,790	Grace Chin-Fa and Tsuming Wu Fellowship	2024-2025
\$500	Pennsylvania Society of Ornithology Research Grant	2023
\$1,000	Hoagy Schaadt & Tom Hardisky Travel Grant	2022
\$500	Commonwealth University Graduate Research Grant	2022
	1ST Place, Graduate Research Poster – COST Research Event	2023
	Howard Jennings Memorial Naturalist Award – Keystone College	2021

Publications

Elias, J.M., Corbin, C.E. Microplastics and Terrestrial Birds: A Review of Plastic Ingestion in Ecological Linchpins. *Journal of Ornithology* 166: 1–8 (2025). https://doi.org/10.1007/s10336-024-02226-4

Elias, J.M., Kibelstis, E., Desantis, T., Smith, R., Corbin, C.E. Documenting Northern Saw-Whet Owls Breeding in Mixed-Deciduous Forest. *Northeastern Naturalist* 31(1): 1-12. (2024). https://doi.org/10.1656/045.031.0101

Elias, J.M., Rier, S.T. Vertical Water Column Position Determines Density and Types of Microplastics in Harvey's Lake, the Largest Natural Lake by Volume in Pennsylvania. *Journal of the Pennsylvania Academy of Science* 31 97(1-2): 1–10, (2023). https://doi.org/10.5325/jpennacadscie.97.1-2.0001

Elias, J.M., Stoleson, S.H. Abundant Natural Cavities in a Deciduous Forest Leads to a Lack of Nest-Box Occupancy by Northern Saw-Whet Owls. *Northeastern Naturalist* 28(2): 202-210, (2021). https://doi.org/10.1656/045.028.0209

(In Review) Elias J.M., Corbin, C.E., Smith, R., Rier, S.T. Microplastic Loading in Terrestrial Birds from Non-Lethal Sampling. (2024).

(In Preparation) Elias J.M., Grames, E.M. Optimal site selection to help design ecological monitoring networks. (2025).

Presentations and Posters

Oral Presentation: *Digitizing and annotating a 25-year time series of Orthoptera stridulations* – Regional Meeting of the Entomological Society of America, Saratoga Springs, NY. 2026.

Poster Presentation: *Optimal site selection to help design ecological monitoring networks* – Annual Meeting of the Ecological Society of America, Baltimore, MD. 2025.

Oral Presentation: *Optimizing monitoring network design for ecological gradients* - BGSO Annual Biology Symposium, Binghamton, NY. 2025.

Poster Presentation: *Method for measuring the number of microplastics in landbird fecal samples collected in the field via bird banding* - BU COST Research Event, Bloomsburg, PA. 2023.

Poster Presentation: *Microplastic Accumulation in the Gastrointestinal Tracts of Varying Terrestrial Avian Species* – Annual Meeting of The Wildlife Society, Spokane, WA. 2022.

Poster Presentation: *Vertical microplastic distribution in a lentic system is influenced by depth and algae – a case study on Harvey's Lake (northeastern Pennsylvania)* - BU COST Research Event, Bloomsburg, PA. 2022.

Oral Presentation: *Abundant natural cavities in a deciduous forest leads to a lack of nest box occupancy for Northern Saw-whet Owls* – Annual Meeting for The Wildlife Society, Virtual. 2021.

Mentorship

The Grames Lab "Cricket Crew": Orthoptera bioacoustics annotation and identification. 2026-

- Isabella Clark
- Reilly Johnson
- Catherine Dong
- William Cohen
- Rabia Khan
- Sofia Nastasi

Jennifer Slater (Binghamton University) 2025-2026

- Binghamton University Projects for New Undergraduate Researchers (BUPNUR)
- Senior Thesis: *Morphometric Identification of Noctuid Moths*

Emily Martelliti (Corbin Lab, Commonwealth University of Pennsylvania) 2023

- Microplastic identification and characterization.
- Senior Thesis: *Measuring the number of microplastics in fecal samples collected from Dumetella carolinensis*

Tatum Omlor (Corbin Lab, Commonwealth University of Pennsylvania) 2023

- Microplastic identification and characterization.

Alex Thornton (Keystone College Honors Program) 2023
Monika Buzcek (Keystone College Honors Program) 2022

Teaching

BIOL 431/531 Ecological Modeling 2026

- Teaching Assistant, Binghamton University
- Teacher of Record: Dr. Eliza Grames

ENVI 363 Measuring the Natural World 2025

- Teaching Assistant, Binghamton University
- Teacher of Record: Dr. Amber Churchill

BIOL 301 Ecology 2023

- Teaching Assistant, Commonwealth University of Pennsylvania
- Teacher of Record: Dr. Steven Rier

BIOL 432 Ornithology 2023

- Teaching Assistant, Commonwealth University of Pennsylvania
- Teacher of Record: Dr. Clay Corbin

BIOL 181 Anatomy and Physiology II 2022

- Teaching Assistant, Commonwealth University of Pennsylvania
- Teacher of Record: Dr. Kate Beishline

BIOL 110 Principles of Biology 2022

- Teaching Assistant, Commonwealth University of Pennsylvania
- Teacher of Record: Dr. Evan Houston

BIOL 4120 Conservation Biology 2023

- Adjunct Instructor, Keystone College
- Teacher of Record: Joseph Elias

BIOL 5221 Ornithology 2023

- Adjunct Instructor, Keystone College
- Teacher of Record: Dr. Joseph Iacovazzi

CHEM 1120 General Chemistry I 2021-2023

- Adjunct Instructor, Keystone College
- Teacher of Record: Dr. Michael Selig

BIOL 108Q Maple Sugaring 2022

- Adjunct Instructor, Keystone College
- Teacher of Record: Joseph Elias

CHEM 1125 General Chemistry II 2021-2022

- Adjunct Instructor, Keystone College
- Teacher of Record: Dr. Michael Selig

Professional Experience

SUNY Binghamton University, Grames Lab 2024-

- *Optimal site selection to help design ecological monitoring networks.* Monitoring networks are vital for tracking ecosystem change. However, many monitoring networks have bias introduced at the research design phase which can skew inference. We are developing an algorithmic approach to quantify and account for bias during site selection under geographic and environmental constraints for representative ecological monitoring initiatives.

- *Machine learning methods to measure Lepidoptera in regions with few taxonomic records.* Many regions lack complete moth records, risking lost insights for conservation, evolution, and medicine. We are developing a morphospecies identification technique to use in areas where ground-truth systems (e.g., iNaturalist, AMI System) perform poorly.

- *Long-term trends in Orthoptera using historical bioacoustics data.* Orthoptera are declining faster than other insect groups and play key roles in the environment (e.g., nutrient cycling, trophic support). We are restoring and reanalyzing a historical bioacoustics collection from eastern Texas. We will customize a Convolutional Neural Network (CNN) model for automated detection, and develop the largest consistent bioacoustics dataset for Orthopterans which will be used to track long-term trends in communities and whether these trends are tied to changing environmental conditions.

University of Pittsburgh, Kitzes Lab 2022-2024

- Bioacoustics Technician - used autonomous recording units (ARUs) from public and private lands to passively survey wildlife. Identified recorded wildlife vocalizations to train machine learning classifiers for continuous wildlife surveying.

Commonwealth University of Pennsylvania, Corbin Lab 2022-2023

- Graduate Research Assistant – assisted research on flycatchers, feather load variation in tree swallow nests, closed respirometry of fish, and prepared labs for Anatomy and Physiology.
- *Microplastic Loading in Terrestrial Birds from Non-Lethal Sampling*. We used landbirds as bioindicators to understand how the ecology and morphology of an animal influences microplastic (MP) ingestion in their habitat. We developed a novel non-lethal method to measure MP ingestion during bird banding.

University of Scranton 2021-2023

- Northern Saw-whet Owl Banding Technician – assisted research on capture rate bias between male and female owls. Helped banding, netting, processing, and drawing blood from owls.
- Principal Investigator: Dr. Robert Smith and Mary Bogert

IUP Research Institute/ SUNY Environmental Science & Forestry 2021

- Point Count & Vegetation Technician – assisted in forest surveys, wood thrush nest searching, banding, and tracking using radio telemetry to guide forest resource use and management.
- Principal Investigators: Dr. Cameron Fiss – Larkin (Jeff Larkin, IUP) and Cohen (Jonathan Cohen, SUNY ESF) Labs

USDA Forest Service Northeast Research Laboratory 2020

- Wildlife Field Technician – assisted with Wood turtle surveys to assess breeding populations and age structure in the Allegheny National Forest. Also, assisted with Cerulean warbler and Northern goshawk banding and surveying to aid conservation efforts as well as aided a study on the impact invasive insects pose to food sources for animal communities.
- Principal Investigators: Dr. Scott Stoleson (Cerulean warbler, Wood turtle surveys, migratory bird banding / spotted-wing drosophila project), Dave Brinker (Northen Goshawk Project), and Daniel Roche (migratory bird banding / spotted-wing drosophila project)

Keystone College Environmental Education Institute (KCEEI) 2018-2021

- Maple Sugaring Manager – managed a sugarbush on KCEEI's nature preserve. Duties included tapping trees, boiling syrup, and public education about silviculture/maple production.
- Howard Jennings Nature Preserve Manager – managed trail maintenance, habitat management for grassland and forest species, stream cleanups, and tree plantings.
- Environmental Educator – assisted summer courses in general ecology, aquatic systems, sustainability, and beekeeping.

Conference Organization

Symposium Organizer – *Night Watch on Autopilot: Innovations in Automated Nocturnal Insect Monitoring* 2026

- Annual Meeting of the Entomological Society of America, Portland, OR

Volunteer and Outreach Experience

North American Breeding Bird Survey 2026-

- Route Surveyor

Moth Research Media Highlight 2025

- Media YouTube Link

Northern Saw-whet Owl Community Outreach Event 2021-

- Owl bander and co-presenter

The Wildlife Society Northeast Student Conclave 2025

- Banding volunteer and co-develop of the student workshop.

The Wildlife Society National Chapter 2023-

- Donald H. Rusch Memorial Game Bird Research Scholarship Committee Member

University of Scranton Biology Club 2024

- Visiting scientist – speaker

Tunkhannock Area High School 2024

- Visiting scientist – speaker

Riverside Adventure Company 2023-2024

- General Manager, Outdoor Outfitter & Educator

Keystone College Environmental Education Institute 2023

- Bird Watching and Ecology Program Leader

DCNR Lackawanna State Park “WinterFest” 2021

- Bird Watching Program Leader

PA Fish and Boat Commission, Unassessed Waters Initiative 2021

- Project Coordinator and field leader

PA Game Commission 2019

- Volunteer - Bat survey, habitat management, aquatic plant transfer.
- Volunteer – Mourning dove, Canada geese, and duck banding.